Model Reference Input Shaping Control of a Nonlinear Rotary Crane with Time-Varying Rope Length

Ho Duc Tho, Naoki Uchiyama, and Kazuhiko Terashima

***Abstract*— In this paper, a model reference input shaping (MRIS) control will be formulated for a luffing dynamics of a rotary crane system with a time-varying rope length. The newly established technique is able to completely suppress the residual vibration for a highly nonlinear time-varying system. The basic idea of the proposed approach is to obtain a reference oscillation from a second-order time-invariant linear system, on which the standard input shaping control is applied. The MRIS control law is then designed to exactly match the real vibration of the system with the reference oscillation. By such a means, an exact zero vibration suppression can be achieved for the actual system. The reference (or unshaped) command is parameterized to cope with control objectives other than the vibration suppression. Simulation results are provided to verify the superiority of the MRIS controller over the standard robust input shaping control method.**

I. Introduction

Rotary cranes are extensively used at harbors to transport large payloads—ranging from several hundred kilograms to dozen of tons—one point to another in the workplace. There are three main motions in a rotary crane system: slewing, luffing, and hoisting. The slewing motion rotates the entire base, whereas the luffing movement lowers/rises the boom. The payload is suspended under the boom tip and its height can be adjusted by the hoisting motion. Both slewing and luffing motions are rotational, thus, the rotary crane is usually described in the spherical coordinate system. The rotary crane can also be considered as a robotic arm with two revolute joints and one prismatic joint. Due to the fact that the rope is flexible, the payload will exhibit a residual vibration. This unwanted vibration substantially lengthens the transportation as well as reduces safety of the system in practice. Since the rotary crane is underactuated, the vibration suppression control of this system poses a significant challenge. Therefore, considerable interests were mounting on vibration controls for rotary crane systems. They can be classified into two main classes: feedback and feedforward controls.

Plenty of studies can be found in the feedback control category. In [1], a delayed feedback controller was established to stabilize the sway angles for a ship-mounted crane. Uchiyama [2] formulated a simple linear model of the 3D rotary crane dynamics and afterward, a constant gain partial state feedback controller was designed to robustly stabilize the system for any rope length. Ouyang et al. [3] proposed a nonlinear controller using only slewing motion to suppress both radial and tangential swing angles. Recently, Sun et al. [4] and Lu et al. [5] introduced passivity-based nonlinear controllers for rotary crane systems subject to roll and heave disturbances. Qian et al. [6] reported a nonlinear controller for an offshore boom crane, which comprises an adaptive robust control law and a learning strategy. To sum up, feedback control techniques demonstrated their excellent properties in dealing with uncertainties and disturbances. However, they require a sophisticated and costly sensor structure, which demands a frequent maintenance. This is the reason that the feedback controllers are still not commonly used in practical cranes. In numerous contexts, feedforward controls are preferable in industry due to their simplicity, ease of implementation, and no additional measurement system being required.

Among the feedforward schemes [7]–[11], the input shaping control [12], [13] is the most widely applied method. Its basic idea is to intelligently modulate the original reference command by convolving with a sequence of impulses called input shaper. The input shaper is designed in such a way that the vibration induced by all impulses is self canceled. If the input shaper causes no vibration, its convolution with any reference command also results in a zero vibration for the system. Danielson [14] implemented a non-robust Zero Vibration (ZV) input shaper and two robust ones, including Zero Vibration Derivative (ZVD) and Specified Insensitivity (SI), to a rotary crane with only luffing motion. This study reported that performance of the robust input shapers are better than the ZV input shaper, which is not robust to modeling errors. The same conclusion was made in [15], where a double-pendulum rotary crane was considered. Recently, Newman and Vaughan [16] proposed a specially designed input shaper that can suppress the vibration under non-initial conditions. However, the above-mentioned studies cannot completely suppress vibration for the rotary crane system, that is to say, only a vibration reduction is guaranteed. Furthermore, the time-varying rope length scenario was not yet considered to an elaborate degree since the standard input shaping control, which is originally designed for a second-order time-invariant linear plant, cannot be trivially applied to a time-varying nonlinear system. Those are the problems needing to be overcome.

In order to address the aforementioned challenges, a new model reference input shaping controller for the luffing motion of the rotary crane will be formulated in this paper. The proposed technique can handle a highly nonlinear luffing pendulum dynamics with an arbitrary (known) time-varying rope length. The newly established input shaping control

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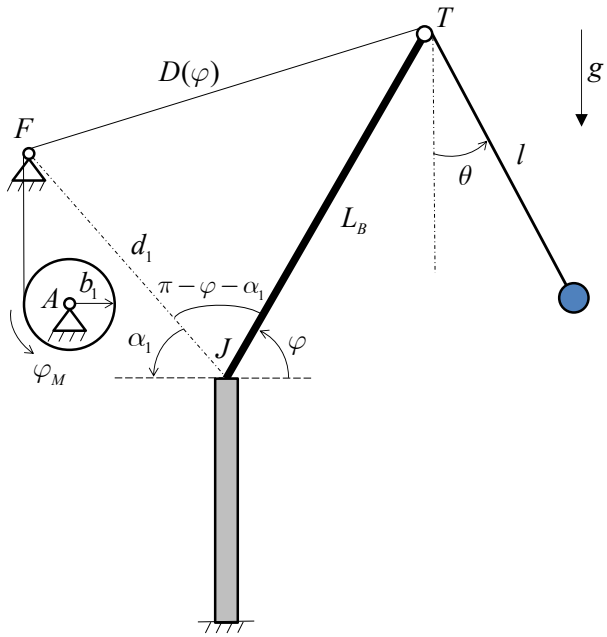

Fig. 1. Mathematical modeling of the luffing dynamics.

TABLE I
SYSTEM PARAMETERS

Parameter	Description	Value
L_B	Boom length (m)	30
d_1	Length of the *FJ* line (m)	9.157
α_1	Auxiliary angle (rad)	1.45
b_1	Radius of the luffing drum (m)	0.3
g	Gravitational acceleration (m/s^2)	9.81

law enables us to use the non-robust ZV input shaper to completely suppress the vibration, thus, the transferring time is significantly faster than standard robust input shapers, which are slow and cannot provide an exact zero vibration cancellation. This is the main contribution of the paper.

II. Mathematical modeling

The swinging motion of the payload induced from the luffing actuation is illustrated in Fig. 1, where $\varphi(t)$, $\theta(t)$, and $l(t)$ are the luffing angle, swing angle, and rope length respectively. In order to vary the luffing angle, a servo motor is attached at the point A to rotate the luffing drum, which connects to the points *F* and *T* by ropes (see Fig. 1). The rotation angle of the luffing motor is denoted as $\varphi_M(t)$. All essential parameters of the system are summarized in Table I.

By using the Euler-Lagrange method, the nonlinear luffing dynamics of the rotary crane system with a time-varying rope length can be described as follows:

$$l\ddot{\theta} + 2\dot{l}\dot{\theta} + g\sin\theta = u(t) \tag{1}$$

where $u(t) = L_B\ddot{\varphi}\sin(\varphi-\theta) + L_B\dot{\varphi}^2\cos(\varphi-\theta)$. The hoisting profile $l(t)$ can be arbitrary. However, $l(t)$ is supposed to be known and $\dot{l}(t)$ is continuous. It is also commonly assumed that the system starts from rest (see [8], [9], [15]), namely $\dot{\varphi}(0) = \dot{l}(0) = \theta(0) = \dot{\theta}(0) = 0$. Furthermore, let $\varphi(0) := \varphi_0$ and $l(0) := l_0$. To obtain a vibration-free transportation, the following terminal conditions must be realized in some finite terminal time τ:

$$\varphi(\tau) = \varphi_f,\ l(\tau) = l_f,\ \dot{\varphi}(\tau) = \dot{l}(\tau) = \theta(\tau) = \dot{\theta}(\tau) = 0. \tag{2}$$

In (2), φ_f and l_f are positive reference values of the luffing angle and the rope length respectively.

The actuator dynamics from the luffing motor to the boom's luffing angle will be presented next. Consider the triangle *FTJ* in Fig. 1, we have

$$D^2(\varphi) = d_1^2 + L_B^2 + 2d_1L_B\cos(\alpha_1+\varphi). \tag{3}$$

Denote the initial angle of the luffing motor as $\varphi_{M0} := \varphi_M(0)$. The initial value D_0 of $D(\varphi)$ can be calculated by replacing φ by φ_0 in (3). Since the variation of the rope length is constant, it can be obtained that

$$(\varphi_M - \varphi_{M0})\,b_1 = D_0 - D(\varphi). \tag{4}$$

Let $\phi := \alpha_1 + \varphi$. From (4), it can be inferred that

$$\varphi_M = \frac{D_0 + b_1\varphi_{M0} - \sqrt{d_1^2 + L_B^2 + 2d_1L_B\cos\phi}}{b_1}. \tag{5}$$

Differentiating (3) and (4), we have

$$\dot{D}(\varphi) = -\frac{d_1L_B\sin\phi}{D}\dot{\varphi} \tag{6}$$

$$\dot{D}(\varphi) = -b_1\dot{\varphi}_M. \tag{7}$$

Comparing (6) and (7) yields

$$\dot{\varphi}_M = \frac{d_1L_B}{b_1}\frac{\sin\phi}{D}\dot{\varphi}. \tag{8}$$

By taking the derivative of (8), it results in

$$\ddot{\varphi}_M = \frac{d_1L_B}{b_1}\left[\frac{D\left(\ddot{\varphi}\sin\phi + \dot{\varphi}^2\cos\phi\right) - \dot{\varphi}\dot{D}\sin\phi}{D^2}\right]. \tag{9}$$

The MRIS control will be designed based on (1) where the control input is $u(t)$. Afterward, it can be translated to the real signal commanding the luffing servo motor in the position or velocity level using (5) or (8) respectively.

The control objective is to design a MRIS control law to bring the system from its initial conditions to the terminal conditions (2) in a minimum time, considering the following actuator constraints:

$$|\dot{\varphi}_M| \le \dot{\varphi}_M^{\max}, |\ddot{\varphi}_M| \le \ddot{\varphi}_M^{\max}, |\dot{l}| \le \dot{l}^{\max}, |\ddot{l}| \le \ddot{l}^{\max}. \tag{10}$$

For a real-size rotary crane system, $\dot{\varphi}_M^{\max} = 5.47$ rad/s, $\ddot{\varphi}_M^{\max} = 2.73$ rad/s^2, $\dot{l}^{\max} = 1.5$ m/s, and $\ddot{l}^{\max} = 0.6$ m/s^2. The following section will be devoted to the design of the MRIS controller.

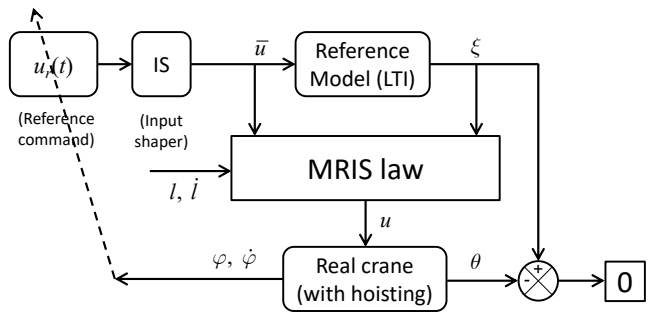

Fig. 2. Basic concept of the MRIS control.

III. MRIS CONTROLLER DESIGN

A. Design concept

The input shaping control is originally established for a second-order time-invariant linear system. A direct application of the standard input shaping control for a time-varying nonlinear system often results in a poor vibration suppression performance. In order to circumvent this problem, the standard input shaping control law is firstly applied to a reference model, which is a second-order time-invariant linear system. Thus, the oscillation of the reference model (named reference oscillation) can be completely suppressed. The actual control input of the time-varying nonlinear system is then derived to exactly match the vibration of the real system with the reference oscillation. By such a means, a complete vibration suppression can be achieved for the real plant. The aforementioned concept of the MRIS scheme is illustrated in Fig. 2. To accomplish the MRIS controller design, the reference command $u_r(t)$ (see Fig. 2) must be determined to fulfill control objectives other than the vibration suppression. This can be done by parameterizing $u_r(t)$ in such a way that the number of unknowns in $u_r(t)$ should be (at least) equal to the number of remaining control targets. For instance, $u_r(t)$ should have two to-be-determined variables to realize two remaining objectives $\varphi(\tau) = \varphi_f$ and $\dot{\varphi}(\tau) = 0$ in (2).

B. MRIS control law

The reference model is introduced as follows:

$$\bar{l}\ddot{\xi} + g\xi = \bar{u}(t) \tag{11}$$

where $\bar{l}$ is a constant nominal rope length. The natural period of (11) is $T = 2\pi\sqrt{\bar{l}/g}$. The control input $\bar{u}(t)$ (see Fig. 2) is designed based on the standard input shaping technique

$$\bar{u}(t) = u_r(t) * I(t) \tag{12}$$

where $*$ is the convolution operator and $I(t)$ is the ZV input shaper, which has the following expression in time domain [17], [18]:

$$I(t) = 0.5\left[\delta(t) + \delta\left(t - T/2\right)\right]. \tag{13}$$

In (13), $\delta(.)$ is the Dirac delta generalized function. As discussed earlier at the end of section III-A, the reference command should have two unknowns. Therefore, it is chosen to be the following bang-bang profile:

$$u_r(t) = \begin{cases} a, & 0 \le t \le t_1 \\ -a, & t_1 < t \le t_f \\ 0, & t > t_f \end{cases} \tag{14}$$

where a and t_1 are to-be-determined variables, and t_f will be gradually reduced to obtain a minimum transferring time that meets the actuator limits (10) (see [8], [10], [19]).

Under the action of the control input (12), the reference oscillation is suppressed, namely

$$\xi\left(t_f + T/2\right) = \dot{\xi}\left(t_f + T/2\right) = 0. \tag{15}$$

The next step is to design the real control input $u(t)$ (see Fig. 2) to match the actual vibration $\theta(t)$ with the reference oscillation $\xi(t)$, that is to say

$$\theta(t) \equiv \xi(t),\ \dot{\theta}(t) \equiv \dot{\xi}(t),\ \ddot{\theta}(t) \equiv \ddot{\xi}(t),\ \forall t \ge 0. \tag{16}$$

By using (1) and (11), we have

$$\begin{aligned} \ddot{\xi} &= -\frac{g}{\bar{l}}\xi + \frac{\bar{u}(t)}{\bar{l}} \\ \ddot{\theta} &= -2\frac{\dot{l}}{l}\dot{\theta} - \frac{g}{l}\sin\theta + \frac{u(t)}{l}. \end{aligned} \tag{17}$$

Applying (16) on (17) results in

$$-2\frac{\dot{l}}{l}\dot{\xi} - \frac{g}{l}\sin\xi + \frac{u(t)}{l} = -\frac{g}{\bar{l}}\xi + \frac{\bar{u}(t)}{\bar{l}} \tag{18}$$

from which the MRIS law $u(t)$ can be extracted as follows:

$$u(t) = \bar{u}(t)\frac{l}{\bar{l}} + g\left(\sin\xi - \frac{l}{\bar{l}}\xi\right) + 2\dot{l}\dot{\xi}. \tag{19}$$

By using the MRIS control law (19), the vibration of the time-varying nonlinear system (1) is completely suppressed, namely

$$\theta\left(t_f + T/2\right) = \dot{\theta}\left(t_f + T/2\right) = 0. \tag{20}$$

Therefore, to accomplish the MRIS controller design, one needs to determine two unknowns a and t_1 in $u_r(t)$ to fulfill the remaining requirements, those are

$$\varphi\left(t_f + T/2\right) = \varphi_f,\ \dot{\varphi}\left(t_f + T/2\right) = 0. \tag{21}$$

Comparing the definition of $u(t)$ in (1) with (19), we have

$$\begin{aligned} &L_B\ddot{\varphi}\sin\left(\varphi - \xi\right) + L_B\dot{\varphi}^2\cos\left(\varphi - \xi\right) \\ &= \left[u_r(t) * I(t)\right]\frac{l}{\bar{l}} + g\left(\sin\xi - \frac{l}{\bar{l}}\xi\right) + 2\dot{l}\dot{\xi}. \end{aligned} \tag{22}$$

Two unknowns a and t_1 can be designed based on (22) to realize (21). However, since (22) is a nonlinear differential-integral equation, an analytical solution may not exist. For this reason, the following optimization problem is configured to determine a and t_1:

$$\begin{aligned} \min_{a,t_1} \quad & \omega_1\left[\varphi\left(t_f + T/2\right) - \varphi_f\right]^2 + \omega_2\dot{\varphi}^2\left(t_f + T/2\right) \\ \text{s.t.} \quad & \text{Nonlinear differential-integral equation (22).} \end{aligned} \tag{23}$$

The weighting coefficients of (23) are $\omega_1 = 10^3$ and $\omega_2 = 10^5$. The nonlinear program (23) can be solved by the *fmincon* routine of MATLAB.

Note that, for any nonlinear optimization solver, good initial guesses of decisive variables are crucial to ensure that the converged (local) minimum has an acceptable quality. The following subsection will be devoted to formulating the initial guesses for a and t_1, which are sufficiently close to the true solutions.

C. Computation of initial guesses for unknown variables

Denote the initial guess of a and t_1 as a_0 and t_{10} respectively. The purpose is to produce reasonable values of a_0 and t_{10} in closed-form. Note that, it is infeasible to obtain an analytical solution of a and t_1 when a full time-varying nonlinear system is used. Therefore, the computation of a_0 and t_{10} will be conducted on a simplified model. This model utilizes the constant nominal rope length $\bar{l}$ and also assumes the swing angle is sufficiently small such that $\sin\theta \approx \theta$ and $\theta \ll \varphi$. Under these assumptions, based on (1), the simplified dynamics of the system is given by

$$\bar{l}\ddot{\theta} + g\theta = \psi(t) \tag{24}$$

where $\psi(t) = L_B\ddot{\varphi}\sin\varphi + L_B\dot{\varphi}^2\cos\varphi$. The initial conditions of (24) are

$$\varphi(0) = \varphi_0,\ \dot{\varphi}(0) = 0,\ \theta(0) = 0,\ \dot{\theta}(0) = 0. \tag{25}$$

The present goal is to design the input $\psi(t)$ of the simplified model (24) in the form of a standard input shaping control law, that is

$$\psi(t) = \psi_r(t) * I(t) \tag{26}$$

where $I(t)$ is the ZV input shaper in (13) and $\psi_r(t)$ is a bang-bang profile similar to (14), that is to say

$$\psi_r(t) = \begin{cases} a_0, & 0 \le t \le t_{10} \\ -a_0, & t_{10} < t \le t_f \\ 0, & t > t_f \end{cases}. \tag{27}$$

Note that, the initial guesses a_0 and t_{10} are present in (27) and their values must be determined to fulfill the conditions $\varphi(t_f+T/2) = \varphi_f$ and $\dot{\varphi}(t_f+T/2) = 0$ in (21). Let us denote $z := -L_B\cos\varphi + L_B\cos\varphi_0$. The terminal conditions (21) can be translated to the z-coordinate as follows:

$$z\left(t_f + T/2\right) = z_f,\ \dot{z}\left(t_f + T/2\right) = 0 \tag{28}$$

where $z_f = -L_B\cos\varphi_f + L_B\cos\varphi_0$. Taking the first and second derivatives of z results in

$$\dot{z} = L_B\dot{\varphi}\sin\varphi \tag{29}$$

$$\ddot{z} = L_B\ddot{\varphi}\sin\varphi + L_B\dot{\varphi}^2\cos\varphi. \tag{30}$$

From (30), we have $\psi(t) = \ddot{z}(t)$. From (26), it yields

$$\ddot{z}(t) = \psi_r(t) * I(t). \tag{31}$$

Note that, unlike (22), (31) is a linear differential-integral equation, which admits an analytical solution. To solve (31), the Laplace transformation method can be used. Taking the Laplace transform for both sides of (31), we have

$$s^2 z(s) = \psi_r(s) I(s). \tag{32}$$

From (27) and (13), the Laplace transforms of $\psi_r(t)$ and $I(t)$ can be easily obtained by

$$\psi_r(s) = -\frac{a_0}{s}\left(2e^{-st_{10}} - e^{-st_f} - 1\right) \tag{33}$$

$$I(s) = 0.5\left(1 + e^{-0.5Ts}\right). \tag{34}$$

Plugging (33) and (34) into (32) and taking the inverse Laplace transform of $z(s)$, we can obtain a closed-form expression of $z(t)$ and $\dot{z}(t)$ in terms of a_0 and t_{10} as shown in (35) at the bottom of this page, where $H(.)$ is the Heaviside function. By using (28) on (35), it can be obtained that

$$a_0 = \frac{4z_f}{t_f^2},\ t_{10} = \frac{t_f}{2}. \tag{36}$$

The values of a_0 and t_{10} in (36) are the initial guesses of a and t_1 when solving the nonlinear program (23).

IV. Simulation results

This section provides simulation results of the proposed MRIS controller and its comparisons with no-control case and the standard robust input shaper ZVDD [12]. In no-control context, both luffing and hoisting motions use trapezoidal velocities to reach their reference values without considering the vibration suppression. The simulations are conducted for the following two cases with different luffing ranges:

$$\begin{aligned} z(t) = &-\frac{1}{2}a_0\Big[\left(t-t_{10}\right)^2 H\left(t-t_{10}\right) + \left(t-t_{10}-T/2\right)^2 H\left(t-t_{10}-T/2\right) - 0.5\left(t-t_f\right)^2 H\left(t-t_f\right) \\ &-0.5\left(t-t_f-T/2\right)^2 H\left(t-t_f-T/2\right) - 0.5\left(t-T/2\right)^2 H\left(t-T/2\right) - 0.5t^2\Big]. \\ \dot{z}(t) = &-\frac{1}{2}a_0\Big[2\left(t-t_{10}\right)H\left(t-t_{10}\right) + \left(t-t_{10}\right)^2\delta\left(t-t_{10}\right) + 2\left(t-t_{10}-T/2\right)H\left(t-t_{10}-T/2\right) \\ &+\left(t-t_{10}-T/2\right)^2\delta\left(t-t_{10}-T/2\right) - \left(t-t_f\right)H\left(t-t_f\right) - 0.5\left(t-t_f\right)^2\delta\left(t-t_f\right) \\ &-\left(t-t_f-T/2\right)H\left(t-t_f-T/2\right) - 0.5\left(t-t_f-T/2\right)^2\delta\left(t-t_f-T/2\right) \\ &-\left(t-T/2\right)H\left(t-T/2\right) - 0.5\left(t-T/2\right)^2\delta\left(t-T/2\right) - t\Big]. \end{aligned} \tag{35}$$

- Case 1: $\varphi_0 = 0.38$ rad (22 degs), $\varphi_f = 0.79$ rad (45 degs), $l_0 = 30$ m, and $l_f = 10$ m.
- Case 2: $\varphi_0 = 0.38$ rad (22 degs), $\varphi_f = 1.22$ rad (70 degs), $l_0 = 30$ m, and $l_f = 10$ m.

The nominal rope length is chosen as $\bar{l} = 0.5\,(l_0 + l_f)$. Note that, t_f is gradually reduced by the bisection method to obtain a minimum value $(t_f)_{\min}$ for which the actuator limitations (10) are met [8], [10]. The computational results are summarized as follows:

- Case 1: $a = 0.206$, $t_1 = 5.553$ s, $(t_f)_{\min} = 11.35$ s.
- Case 2: $a = 0.550$, $t_1 = 5.552$ s, $(t_f)_{\min} = 11.35$ s.

The computational times of Case 1 and Case 2 are 10.76 s and 9.53 s respectively on a personal computer with 2.7-GHz Intel Core i5-4310M and 8 GB of RAM. It is remarked that the total transferring time of the MRIS controller is $(t_f)_{\min} + T/2$, where $T/2$ is the delay caused by the ZV input shaper. In two cases, an identical minimum transportation time is obtained due to the actuator limitations of the hoisting motion. It is also important to note that, when initial guesses of a and t_1 are both set to zero, the optimization solver (i.e., *fmincon*) fails to provide a good solution and sometimes, infeasibility is encountered. Thus, the appropriate guesses suggested in section III-C are crucial.

A standard robust ZVDD input shaper is employed to compare with the proposed MRIS controller. The formula of the ZVDD input shaper in time domain is given by [17]

$$I_1(t) = \frac{1}{8}\delta(t) + \frac{3}{8}\delta\left(t - \frac{T}{2}\right) + \frac{3}{8}\delta\left(t - T\right) + \frac{1}{8}\delta\left(t - \frac{3T}{2}\right). \tag{37}$$

The ZVDD input shaper is directly convoluted with an unshaped luffing acceleration command $\ddot{\varphi}_r(t)$, which takes a similar bang-bang form as in (14). The resultant convolution product $\ddot{\varphi}(t) = \ddot{\varphi}_r(t) * I_1(t)$ is fed to the crane dynamics (1) to obtain the swing response using a numerical integration method (e.g. 4th-order Runge-Kutta). The minimum transfer time under actuator limitation constraints in the ZVDD case is also determined by the bisection method similar to the aforementioned MRIS control.

The comparative simulation results between the MRIS control, a standard robust input shaping control using the ZVDD input shaper (37), and a no-control case are given in Figs. 3–4 for Case 1 and Case 2. Note that, the full time-varying nonlinear system (1) is used in the simulations. From Figs. 3–4, it can be observed that the MRIS control law is able to completely suppress the vibration, whereas there still exists a significant residual oscillation after the motion in the case when the robust ZVDD input shaper is utilized. Furthermore, the control input magnitude of the MRIS controller is considerably smaller than the ZVDD input shaping control. Therefore, it has been shown that, under a relatively similar transportation time, the proposed MRIS scheme is superior than the standard robust input shaping method. This confirms the main aim of the paper.

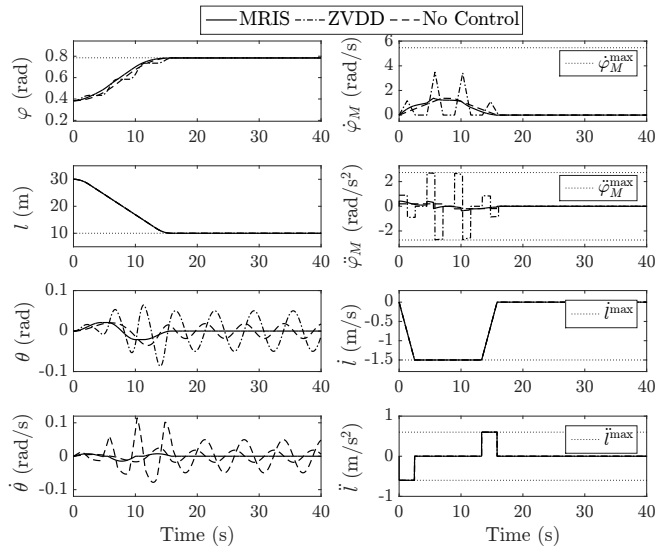

Fig. 3. Comparative simulation result in Case 1.

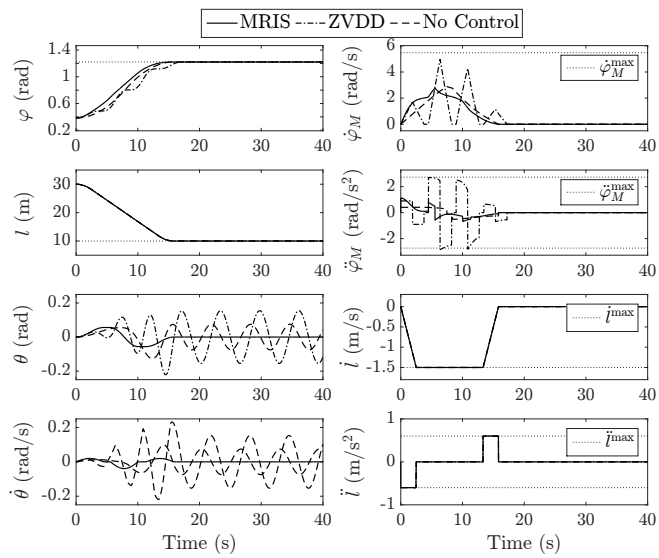

Fig. 4. Comparative simulation result in Case 2.

V. Conclusions and Future Work

This paper established a model reference input shaping controller—a new scheme to obtain an exact zero vibration suppression for a highly nonlinear time-varying system found in the luffing dynamics of the rotary crane. The performance of the MRIS controller was shown to be superior than the standard robust input shaping control using the ZVDD input shaper. In our future work, the above-mentioned results will be extended for a three-dimensional rotary crane system with both luffing and slewing dynamics in the presence of a time-varying rope length. In addition, reducing the computational time will also be addressed in future.

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